VolDynamics

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Context first. Pressure second. Execution last.

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1. Overview

Flow.42 VolDynamics reads the tape and marks the bars where volume, price and aggressor behavior line up into something meaningful. Instead of leaving you to eyeball

every candle, it colors the few bars that tell a story: where a push got absorbed, where a run ran out of fuel, and where price broke an edge with real participation behind it.

Three pattern types sit at the core. **Absorption** is big volume that barely moves price — a large wick, a small body, a close near the opposite end, the fingerprint of hidden supply or demand soaking up aggression. **Exhaustion** is a high-volume rejection at the end of a directional run, the move running low on fuel. **Volume Push** is strong participation expanding near an important price area, which can support a continuation when price follows through or warn of a late push when it fails.

Underneath the patterns is a **volume engine** that learns what normal volume looks like for your instrument — including the time-of-day rhythm, so the lunchtime lull is judged differently from the open — and flags true surges relative to that baseline. An optional **Order Flow** layer then confirms each pattern against buy/sell aggression, so you can demand that the tape agrees before a bar is colored.

Because volume is treated generically as units traded, VolDynamics works across futures, FX and crypto alike. It is a context tool: it highlights effort that matters, but the entry, exit and risk decisions remain yours.

2. What you see on the chart

VolDynamics mainly works by **coloring bars**. Each pattern and surge type has its own color, so a glance tells you what the indicator saw:

- **Absorption** — medium purple. Aggressive flow got absorbed; the wick and close structure suggest the push failed.
- **Exhaustion** — gold. End-of-run behavior; a tail and a smaller body after a directional run.
- **Volume Push (edge break)** — deep sky blue. A range edge breaks after pressure builds in a cluster near the edge.
- **Volume Surge up** — lime green. A high-volume bar (not absorption or exhaustion) closing bullish.
- **Volume Surge down** — indian red. A high-volume bar closing bearish.

When more than one condition matches the same bar, the most structurally significant one wins: an edge break takes priority over absorption or exhaustion, which in turn take priority over a plain volume surge. Every color is configurable, and each pattern can be set to color its bars or leave them unchanged. When pattern tooltips are on, hovering a colored bar explains what was detected.

3. The three patterns

3.1 Absorption

Absorption is the signature of a push being quietly soaked up. A bar qualifies when it shows a large wick relative to its range, a small body, and a close pushed toward the end opposite the wick — the market tried to go one way, met a wall, and closed back the other way. The wick, body and close requirements are all adjustable: **Wick Min Fraction** sets how much wick is required, **Body Max Fraction** caps the body, and **Close Top Fraction** / **Close Bottom Fraction** govern where the close must land for the bullish and bearish cases. Turn the whole pattern on or off with **Enable Absorption**.

3.2 Exhaustion

Exhaustion looks for the same kind of rejection wick, but specifically at the end of a directional run — the tell of a move running out of steam. So in addition to the wick, body and close requirements, it asks for a prior run to have happened: **Lookback Bars** sets how far back the run is measured, and **Min Run (ATRs)** sets how large that run must be, expressed in ATR multiples so it scales with volatility. **Wick Min Fraction**, **Body Max Fraction**, **Close Top Max** and **Close Bottom Min** shape the bar itself.

3.3 Volume Push (edge break)

Volume Push catches a decisive break of a range edge after pressure has been building. It looks back over a **Cluster Window** for a cluster of at least **Min Spike Count** clean surges, checks that the cluster sits near the edge of a swing range (defined by **Range Lookback** and how close counts, set by **Edge Fraction**), and then flags the break. A **Cooldown Bars** setting spaces consecutive signals out so you do not get a burst of them in one spot. This

is the pattern that can mark genuine initiation — or, when price fails to follow through, warn that the push came too late.

4. The volume engine

The volume engine is what lets VolDynamics tell a real surge from ordinary activity. Turned on with **Enable Volume Engine**, it learns a baseline of normal volume for your instrument and compares each bar against it. Crucially, the baseline is **time-of-day aware**: it builds an intraday profile in buckets (set by **Intraday Time Bucket**), so a bar at the busy open is judged against other opens and a quiet midday bar against other midday bars. That keeps the surge calls honest across the session.

How quickly the baseline adapts is yours to set. **Baseline Speed** offers Snappy, Balanced or Smooth presets; **Startup Days for Baseline** controls how much history seeds the initial baseline; and **Baseline Update Speed** fine-tunes how strongly each new bar nudges it. More history and slower updates give a steadier baseline; less history and faster updates make it more reactive.

A surge can be defined two ways. By default it is **relative** — a bar must be at least **Surge Factor vs Normal** times the baseline, bounded by **Min Surge Factor** and **Max Surge Factor** so the threshold stays sensible in both quiet and extreme conditions. Alternatively, **Use Fixed Surge Volume** switches to an **absolute** rule: a bar must trade at least a fixed number of units (**Fixed Surge Volume**), which is handy on instruments where you think in concrete size (for example, ≥ 3 BTC). And if you would rather not choose at all, **Auto-Tune Surge Level** adjusts the threshold automatically based on current volatility.

5. Volume surge display

The engine's output is shown through the **05a Display Volume Surge** group. **Marker** picks the visualization: None (no surge coloring), Spike bars (color only the surge bars), or Volume intensity (color every bar by how its volume compares to the baseline — a quick heatmap-style read of the whole session). **Directional** classifies each surge as bullish or bearish from the price move, and **Confirm Bars** decides how that direction is judged —

comparing close to open, or close to a close several bars back. The two surge colors, **Surge Up Color** and **Surge Down Color**, are configurable.

6. Order Flow confirmation

The patterns above read price and volume. The **Order Flow** layer adds a third lens — who was aggressive, buyers or sellers — and lets you require that the tape agrees before a pattern is accepted. Turn it on with **Enable Order Flow** and choose where buy/sell volume comes from with **Source**: from up/down ticks, or from Bid/Ask classification.

Confirm Mode sets how demanding it is. Off ignores order flow entirely; **Soft** confirms a pattern if any one order-flow criterion is met; **Strict** requires all of them. The criteria themselves are tunable per pattern: **Absorption Min Aggression** sets how one-sided the flow must be for absorption, **Absorption Max Progress / Block** and **Exhaustion Max Progress / Block** cap how far price may travel per block of volume (the block size set by **Progress Volume Block**), and **Volume Push Min Dominance** sets how dominant one side must be on a breakout bar. Strict mode with tight thresholds gives you the cleanest, rarest signals; Soft mode keeps more of them.

7. Markets: futures, FX and crypto

VolDynamics treats volume generically as units traded, so the same logic works on futures (contracts), FX (lots) and crypto (coin size). The **Volume Units** setting tells it how to interpret volume — Native suits most markets, with Notional and Scaled available for cases where you want a different basis. **Volume Unit Multiplier** is an optional convenience, most useful in crypto: a multiplier of 100 turns 1 BTC into 100 units, giving you rounder numbers to reason about. Keep it at 1 for most instruments.

For 24/7 crypto, the time-of-day baseline still applies and the fixed-surge option is often the most intuitive — you simply state the minimum size that counts as a surge. For futures and FX, the relative surge factor or Auto-Tune usually feels most natural.

8. Sessions, tooltips and the help panel

Sessions. VolDynamics can limit its work to time windows — a Pre-Market window and an RTH window, each switchable with its own start and end time. With **Ignore Session Filter** on, the windows are bypassed and bars are evaluated at any time. Times accept both 24-hour and 12-hour formats in your local zone.

Tooltips. Show Pattern Tooltips adds a mouse-over explanation on absorption, exhaustion and volume-push bars, describing the detected pattern — useful while you are learning to read the colors.

Help panel. A shared on-chart help panel keeps the basics at hand. **Show Help Panel** toggles it, **Enable Help Hotkey** lets you open it with the Flow.42 hotkey, and its width, height and opacity are adjustable.

9. Full parameter reference

The settings appear in the NinjaTrader properties panel, grouped as you see them there. "Default" is the value the indicator ships with.

01 Absorption

Setting	Default	What it does
Enable Absorption	On	Turns absorption detection on or off.
Wick Min Fraction	0.40	The minimum wick size, as a fraction of the full candle range, required for absorption.
Body Max Fraction	0.45	The maximum body size, as a fraction of the full range, allowed for absorption.
Close Top Fraction	0.60	For bullish absorption, the close must be at or above this fraction of the high.

Setting	Default	What it does
Close Bottom Fraction	0.40	For bearish absorption, the close must be at or below this fraction of the low.
Absorption Bars	Color	Whether absorption bars are colored or left unchanged. Options: None, Color.
Absorption Color	MediumPurple	The bar color used when absorption is detected.

02 Exhaustion

Setting	Default	What it does
Enable Exhaustion	On	Turns exhaustion detection on or off.
Lookback Bars	8	The number of bars used to measure the prior directional run.
Min Run (ATRs)	1.00	The minimum prior run, in ATR multiples, required to qualify as exhaustion.
Wick Min Fraction	0.45	The minimum wick size, as a fraction of the full range, required for exhaustion.
Body Max Fraction	0.45	The maximum body size, as a fraction of the full range, allowed for exhaustion.
Close Top Max	0.60	For upside exhaustion, the close must not be higher than this fraction of the range.
Close Bottom Min	0.40	For downside exhaustion, the close must not be lower than this fraction of the range.
Exhaustion Bars	Color	Whether exhaustion bars are colored or left unchanged. Options: None, Color.
Exhaustion Color	Gold	The bar color used when exhaustion is detected.

03 Volume Push

Setting	Default	What it does
Enable Volume Push	On	Turns Volume Push (edge break) detection on or off.
Cluster Window (Bars)	12	How many bars back are scanned for the spike cluster before a breakout.
Min Spike Count	2	The minimum number of clean spikes required inside the cluster.
Range Lookback (Bars)	30	The number of bars used to define the swing range around the breakout.
Edge Fraction	0.15	How close the spike cluster must sit to the range high or low to count as an edge.
Cooldown Bars	20	The minimum number of bars to wait between two Volume Push signals.
Volume Push Bars	Color	Whether Volume Push bars are colored or left unchanged. Options: None, Color.
Volume Push Color	DeepSkyBlue	The bar color used when Volume Push is detected.

04 Order Flow

Setting	Default	What it does
Enable Order Flow	On	Turns order-flow confirmation on or off.
Source	BidAsk	Where buy/sell volume is derived from: UpDownTick or BidAsk classification.
Confirm Mode	Soft	Off ignores order flow; Soft confirms if any one criterion holds; Strict requires all criteria.
Absorption Min Aggression	0.60	The minimum one-sided dominance for absorption (0.60 = 60% of volume on one side).

Setting	Default	What it does
Absorption Max Progress / Block	0.30	The maximum price progress (in ticks) per volume block for absorption.
Exhaustion Max Progress / Block	0.35	The maximum price progress per volume block for exhaustion.
Volume Push Min Dominance	0.60	The minimum aggressor dominance on the breakout bar to confirm a Volume Push.
Progress Volume Block	1000	How many volume units make one block for the progress-per-block checks.

05 Volume Dynamics

Setting	Default	What it does
Enable Volume Engine	Off	Turns the volume engine on or off. Works for all markets — volume is contracts, lots or coin size.
Volume Units	Native	How volume is interpreted: Native, Notional or Scaled. Native suits most markets.
Volume Unit Multiplier	1.0	An optional multiplier, useful in crypto (e.g. 100 turns 1 BTC into 100 units). Keep at 1 for most instruments.
Baseline Speed	Balanced	How fast the normal-volume baseline adapts: Snappy, Balanced or Smooth.
Startup Days for Baseline	5	How many past days seed the initial baseline. More days are smoother; fewer are more reactive.
Intraday Time Bucket (min)	1	The size of each time bucket for the intraday volume profile. 1–5 is recommended for futures and crypto.
Baseline Update Speed	0.05	How strongly each new bar updates the baseline. Lower is smoother; higher is faster.
Use Fixed Surge Volume	On	If on, surges require a fixed minimum volume. If off (or Auto-Tune on), surges are relative to normal.

Setting	Default	What it does
Fixed Surge Volume (units)	1000	The minimum volume for a surge in fixed mode. Units depend on the instrument (e.g. 3.0 on BTC = ≥ 3 BTC).
Surge Factor vs Normal	2.0	How many times above the baseline a bar must be to count as a surge (relative mode).
Min Surge Factor	1.2	The lowest allowed surge factor, to avoid oversensitivity in quiet periods.
Max Surge Factor	6.0	The highest allowed surge factor, to prevent unrealistic thresholds in extreme volume.
Auto-Tune Surge Level	Off	Automatically adjusts the surge threshold based on current volatility.

05a Display Volume Surge

Setting	Default	What it does
Marker	Spike bars	The visualization: None, Spike bars (surges only), or Volume intensity (every bar vs baseline).
Directional	On	Classifies surges as bullish or bearish from the price move.
Confirm Bars	2	0 compares close to open; otherwise compares close to the close that many bars back.
Surge Up Color	LimeGreen	The bar color for bullish volume surges.
Surge Down Color	IndianRed	The bar color for bearish volume surges.

06 Session Settings

Setting	Default	What it does
Ignore Session Filter	Off	When on, no session gating is applied and signals can occur at any time.
Use Pre-Market	On	Enables the Pre-Market window.
Pre Market Start (HH:mm)	09:00	The Pre-Market start time, in your local time. Accepts 24-hour and 12-hour formats.
Pre Market End (HH:mm)	13:00	The Pre-Market end time, in your local time.
Use RTH	On	Enables the Regular Trading Hours window.
RTH Start (HH:mm)	15:00	The RTH start time, in your local time.
RTH End (HH:mm)	22:00	The RTH end time, in your local time.

07 Tooltips

Setting	Default	What it does
Show Pattern Tooltips	Off	Shows a mouse-over tooltip on absorption, exhaustion and volume-push bars explaining the detected pattern.

Help & Hotkey

Setting	Default	What it does
Show Help Panel	—	Shows or hides the shared Flow.42 help panel on this chart.
Enable Help Hotkey	—	Allows toggling the help panel with the global Flow.42 hotkey.
Assigned Help Hotkey	CTRL + H	The shortcut that toggles the help panel. Read-only.
Help Panel Width (px)	—	The preferred width of the help panel in pixels.

Setting	Default	What it does
Help Panel Height (px)	—	The preferred height of the help panel in pixels.
Help Panel Opacity (%)	—	The opacity of the help panel, from 0 to 100.

10. Suggested starting settings

Out of the box, the three patterns and the order-flow layer are on, and the volume engine is off — a good way to first see where absorption, exhaustion and volume pushes land on your instrument. Watch the colored bars for a few sessions before changing anything.

When you want surge coloring too, turn on **Enable Volume Engine**. On futures and FX, leave the surge in relative mode (or switch on Auto-Tune) and let the baseline learn — give it the few startup days it needs. On crypto, the fixed-surge option is often clearest: state the minimum coin size that counts as a surge and go from there. If the whole-session heatmap interests you, set Marker to Volume intensity to color every bar by its volume versus the baseline.

For confirmation, start in Soft mode so the order-flow layer trims the weakest patterns without removing too many. Move to Strict only once you know the patterns well and want the cleanest, rarest signals. As a rule, the most instrument-sensitive settings are the surge definition (fixed volume, surge factor, or Auto-Tune) and the order-flow thresholds — tune those first.

11. Troubleshooting

No bars are colored. Confirm you are inside a session window or turn on Ignore Session Filter, and check that the patterns you expect are enabled. The indicator needs a short warm-up of bars before it begins coloring.

No volume surges appear. The volume engine is off by default — turn on Enable Volume Engine. If it is on, the baseline needs its startup days to learn; very early in that window,

surge calls may be sparse. In fixed mode, make sure Fixed Surge Volume is reachable for your instrument.

Too many or too few surges. In relative mode, raise or lower Surge Factor vs Normal (bounded by the min and max). In fixed mode, adjust Fixed Surge Volume. Or switch on Auto-Tune to let volatility set the level.

Order flow seems to remove everything. Strict mode requires every order-flow criterion at once. Switch to Soft, or loosen the per-pattern thresholds (aggression, progress-per-block, dominance). Also confirm your data feed supports your chosen Source — Bid/Ask classification needs the right market data.

Colors collide on a bar. When several conditions match, the most significant wins — edge break over absorption/exhaustion over a plain surge. That is by design, so a bar shows its strongest read.

Crypto numbers feel awkward. Use the Volume Unit Multiplier to scale coin size into rounder units, and consider fixed surge volume so you are reasoning in concrete size.

12. Glossary

Absorption — big volume that barely moves price: a large wick, small body and close near the opposite end, suggesting hidden supply or demand soaking up aggression.

Exhaustion — a high-volume rejection at the end of a directional run, suggesting the move is running out of fuel.

Volume Push (edge break) — a decisive break of a range edge after a cluster of surges builds near that edge.

Volume surge — a bar whose volume is unusually high relative to the learned baseline.

Baseline — the engine's evolving picture of normal volume, including the time-of-day rhythm of the session.

Surge factor — how many times above the baseline a bar must trade to count as a surge (relative mode).

Order flow — the read of which side, buyers or sellers, was the aggressor, used to confirm patterns.

Confirm mode — how strict order-flow confirmation is: Off, Soft (any criterion), or Strict (all criteria).

Volume units — how volume is interpreted for the instrument: contracts, lots or coin size.

Flow.42 VolDynamics is a context and visualization tool. It does not provide automated trade signals or trading advice. Trade less. Trade better. Trade with clarity.

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